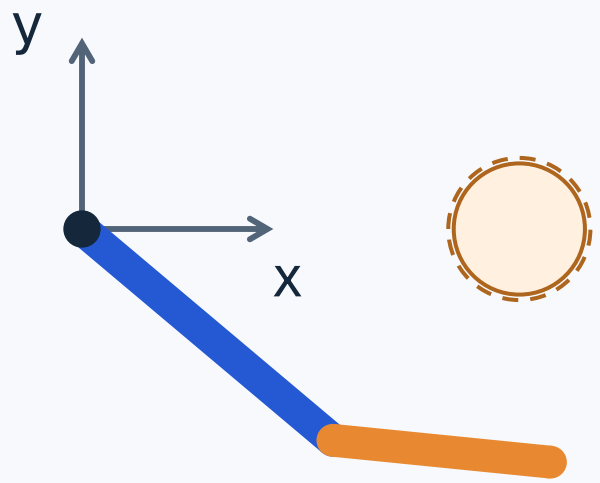
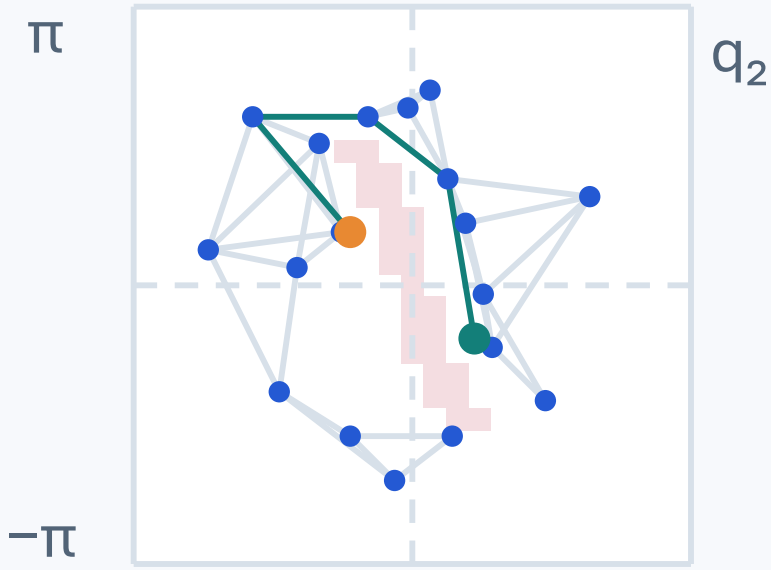
Configuration space & PRM

Describe the whole robot. Find a collision-free route.



20 min concepts · 25 min demo

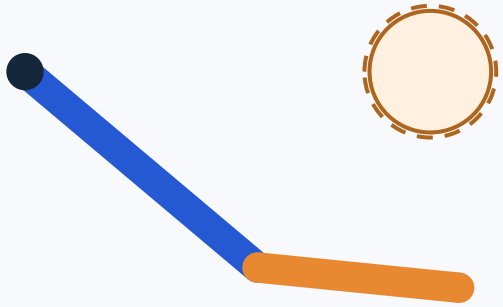


Illustrative roadmap

Why motion planning?

A reachable goal does not guarantee a clear route.

Start: free



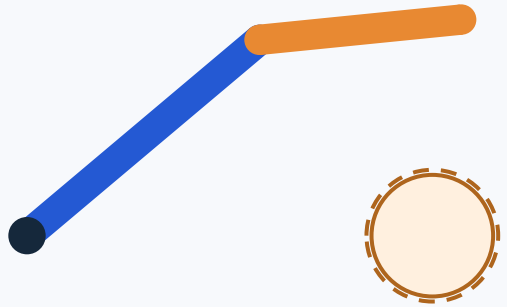
q = (−0.7, 0.6)

Midpoint: collision



q = (0, 0)

Goal: free



q = (0.7, −0.6)

Three kinds of planning

Choose actions, search connections, and realize robot motion.

Task planning

Approach → grasp → place
Choose actions.

Graph search

Find a route among known connections.

Motion planning

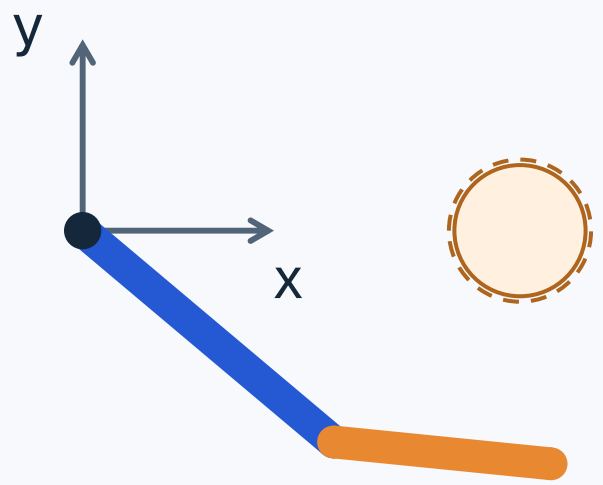
Move the whole robot without collision.

PRM builds geometric connections, then searches them.

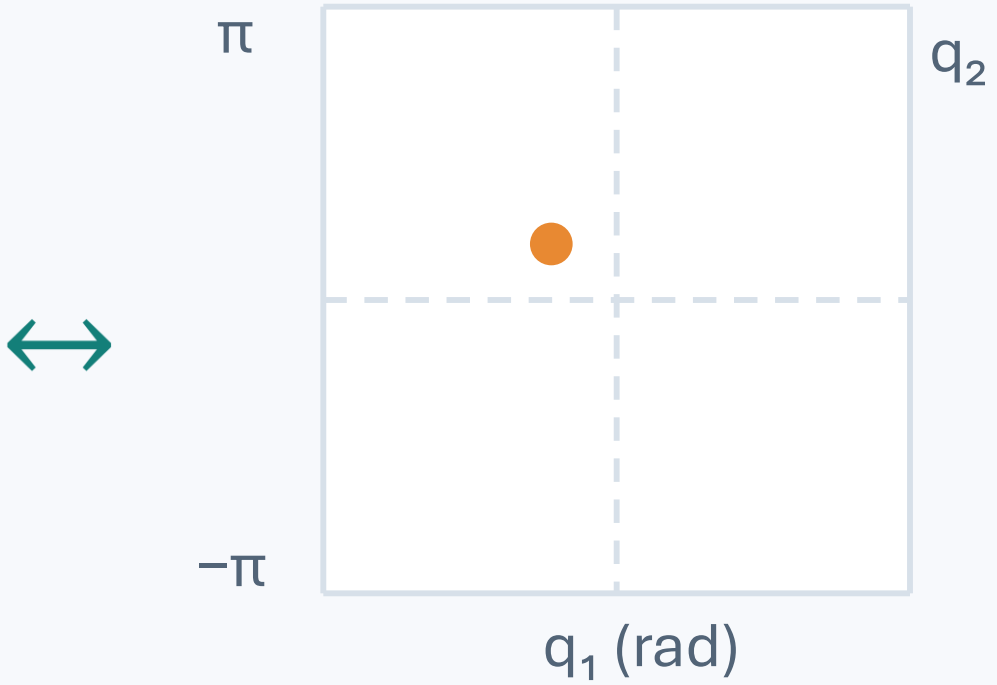
Workspace versus configuration space

One point in configuration space represents the whole arm.

Workspace · metres



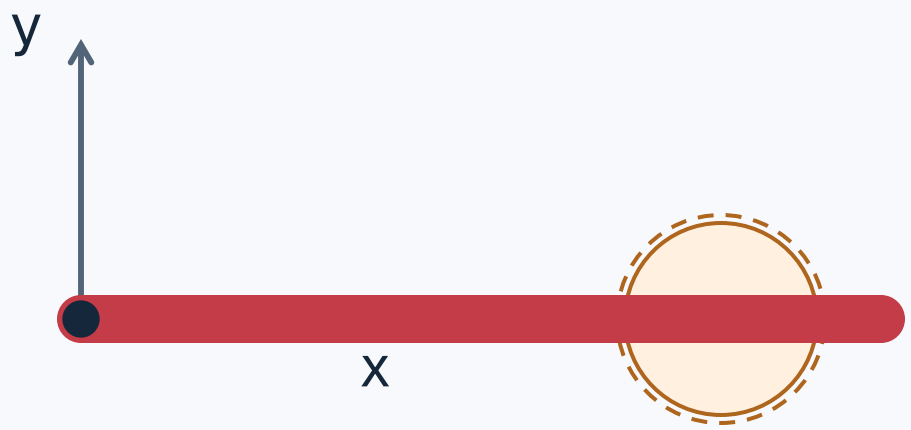
Links: 0.6 m + 0.4 m



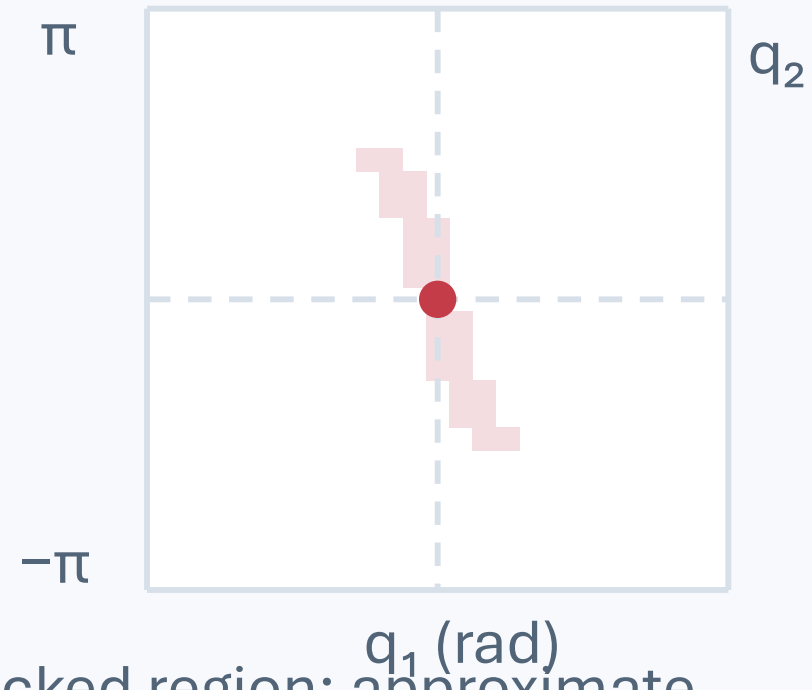
q = (−0.7, 0.6) rad

Obstacles become invalid configurations

Check the robot geometry, not only its end effector.



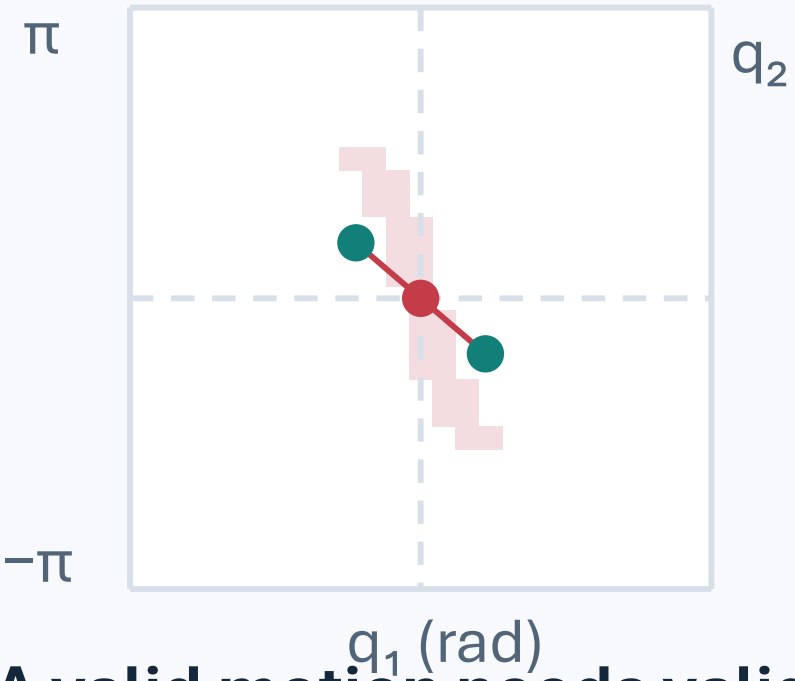
Tip is clear. Link 2 collides.



Blocked region: approximate

Safe endpoints, blocked motion

A valid vertex does not make every outgoing motion valid.



$$q(s) = (1-s)q_{start} + sq_{goal}$$

s = 0, 1: free

s = ½: collision

A valid motion needs valid intermediate configurations.

PRM: build, query, verify

The live demo will make these steps clearer.

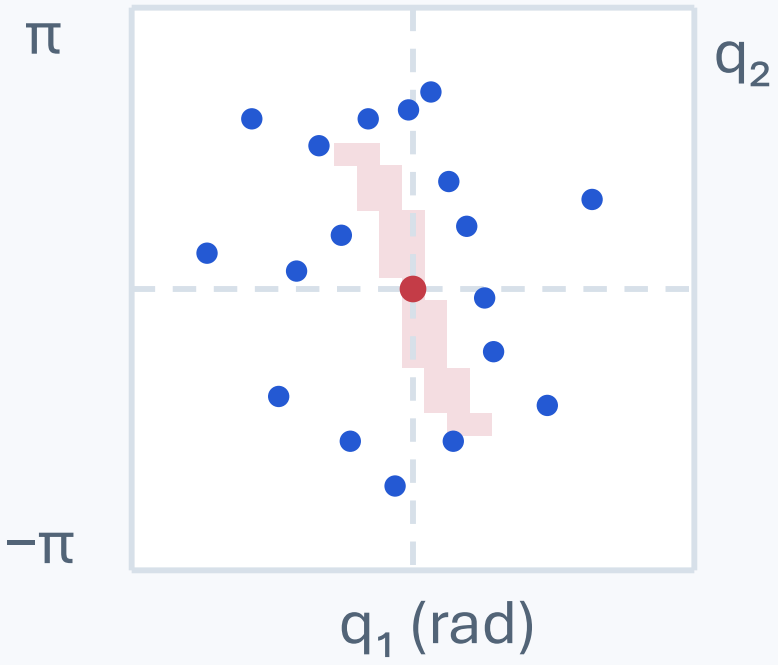
1 **Sample** → 2 **Filter** → 3 **Neighbors** → 4 **Validate**

5 **Store graph** → 6 **Attach query** → 7 **Search** → 8 **Verify**

Build once → reuse for many start–goal queries.

1 · Sample configurations

Draw joint-angle pairs inside the allowed configuration space.



Attempts ≠ accepted samples

Implement

Seeded uniform angles; bounded attempts.

Check

Limits, repeatability, attempt exhaustion.

2 · Keep free configurations

Forward kinematics supplies the geometry for collision checks.



$$\text{clearance}(q) > 0.01\ \text{m}$$

Thickness + padding matter.

Implement

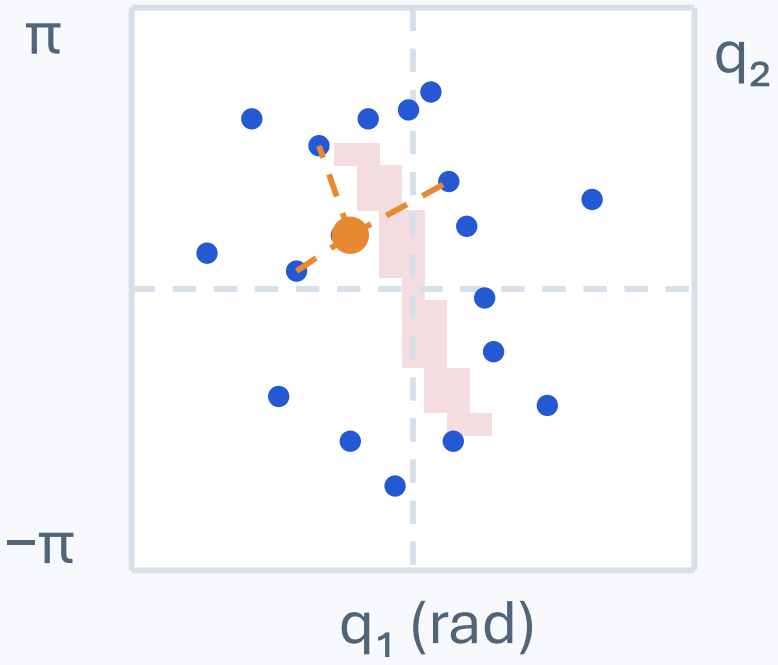
FK → both link capsules → clearance.

Check

Either link, contact, and known poses.

3 · Choose nearby candidates

Distance proposes a connection; geometry decides whether to keep it.



$$d(q,a) = \|q-a\|_2$$

Implement

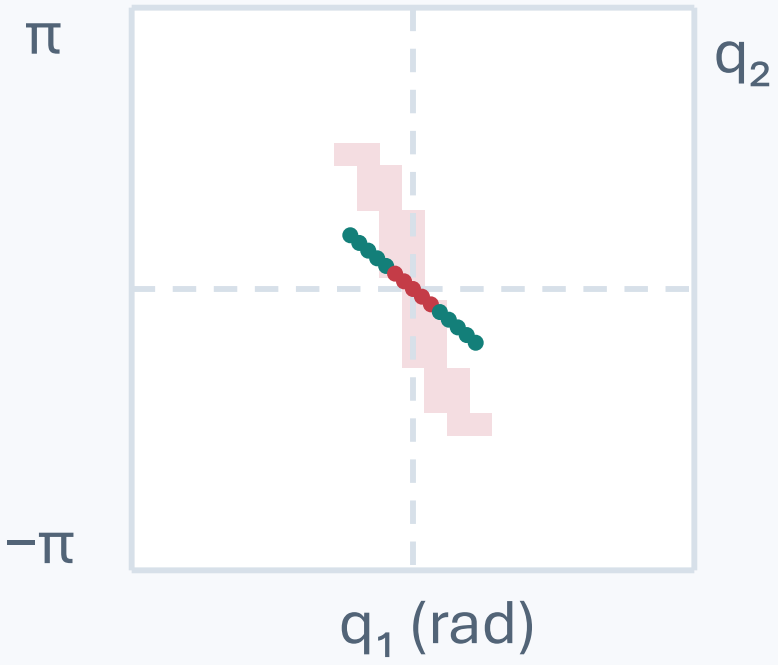
Sort joint-space distance; exclude self.

Check

Known ordering; fewer than k vertices.

4 · Validate local motions

Inspect configurations along the connection, including endpoints.



≤ 0.01 rad per joint per step

Implement

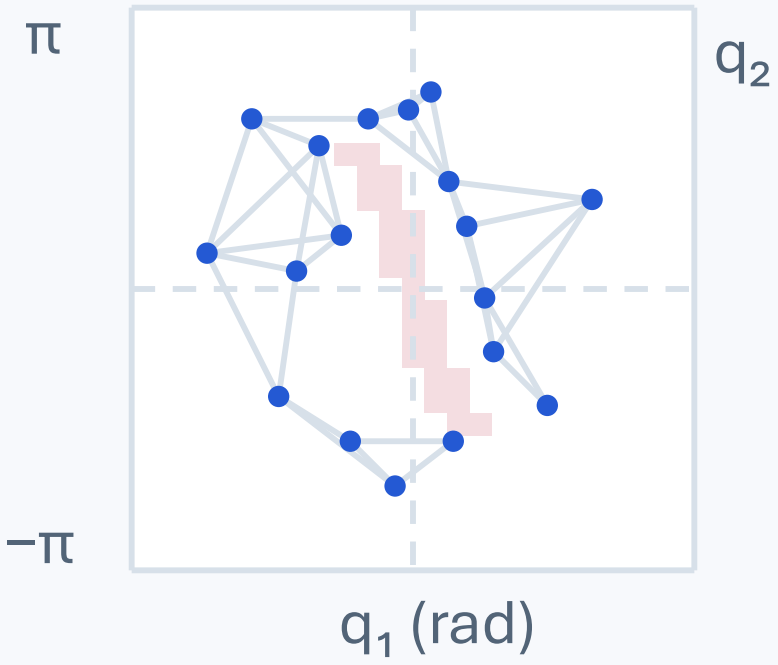
Interpolate; check every sampled pose.

Check

Blocked midpoint; endpoint inclusion.

5 · Store the roadmap

Configurations become vertices; validated motions become edges.



$$w_{ab} = \|q_a - q_b\|_2$$

Implement

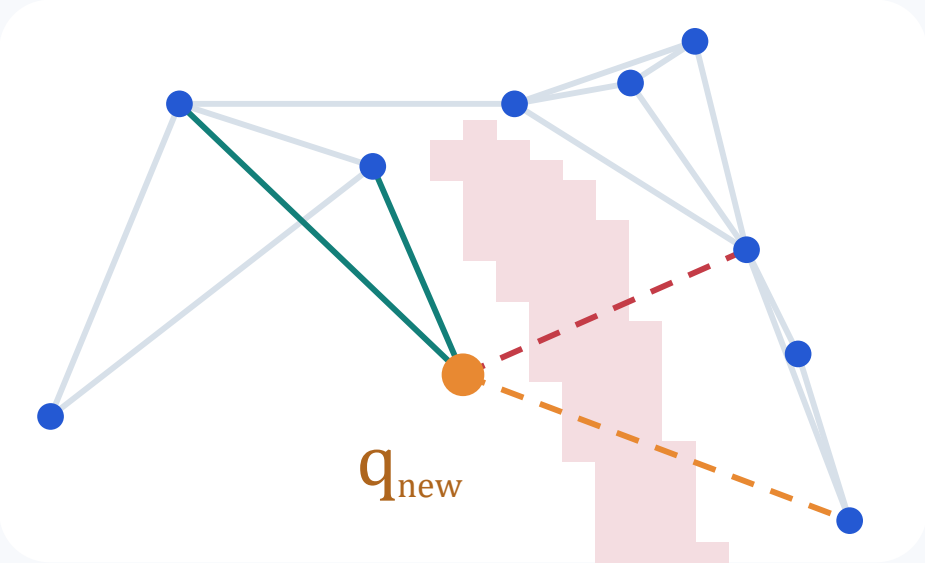
Undirected adjacency; length weights.

Check

Symmetry, costs, no duplicate edges.

6 · Use an unsampled point

Add the actual start or goal; connect it without snapping.



— valid - - rejected

Orange dashed: candidate

Implement

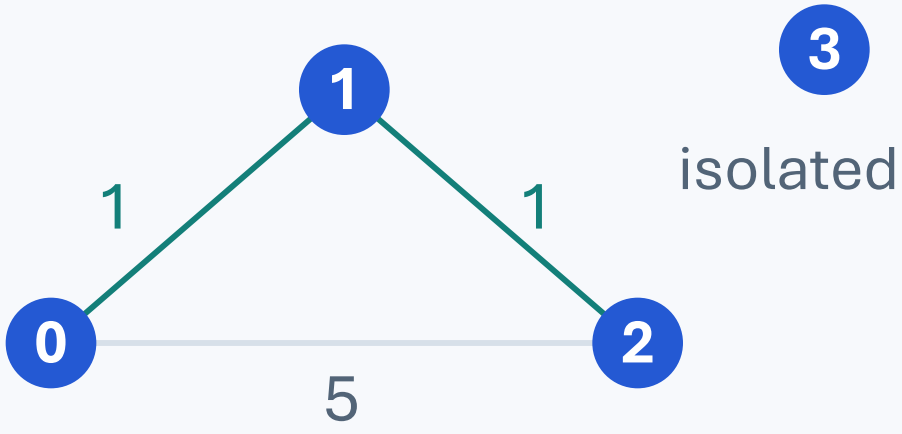
Validate endpoint; copy; add; connect.

Check

Exact endpoint retained; original unchanged.

7 · Search the graph

Dijkstra finds a minimum-cost route among the validated edges.



0 → 1 → 2: total cost 2

Implement

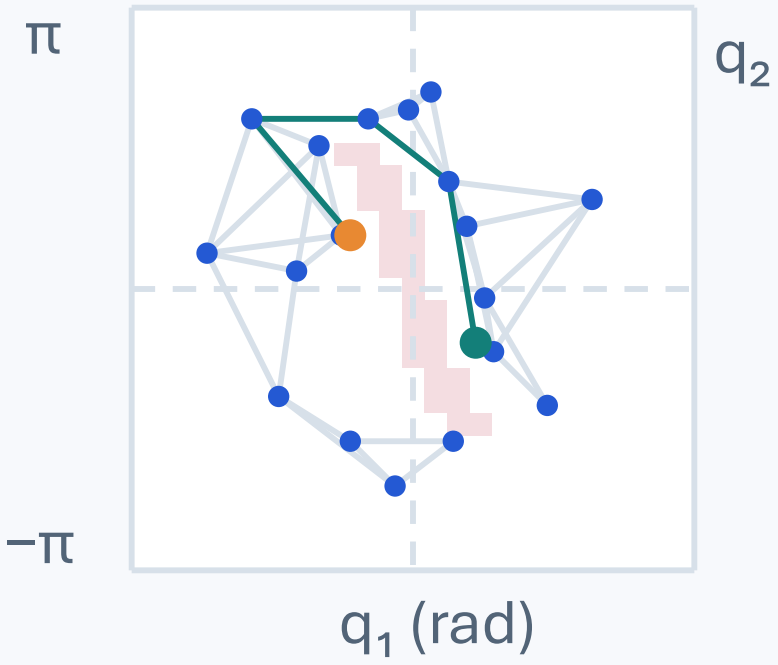
Heap frontier + costs + parent links.

Check

Known shortest route; disconnected graph.

8 · Verify the returned path

Check endpoints, segments, and cost before interpreting playback.



Replay more densely: 0.002 rad

Implement

Replay segments; sum joint-space lengths.

Check

Endpoints, clearance, and reported cost.

Interpret success and failure

A finite roadmap can miss a route that physically exists.

20 samples · seed 7

91 edges
Path length: 5.630 rad

300 samples · seed 7

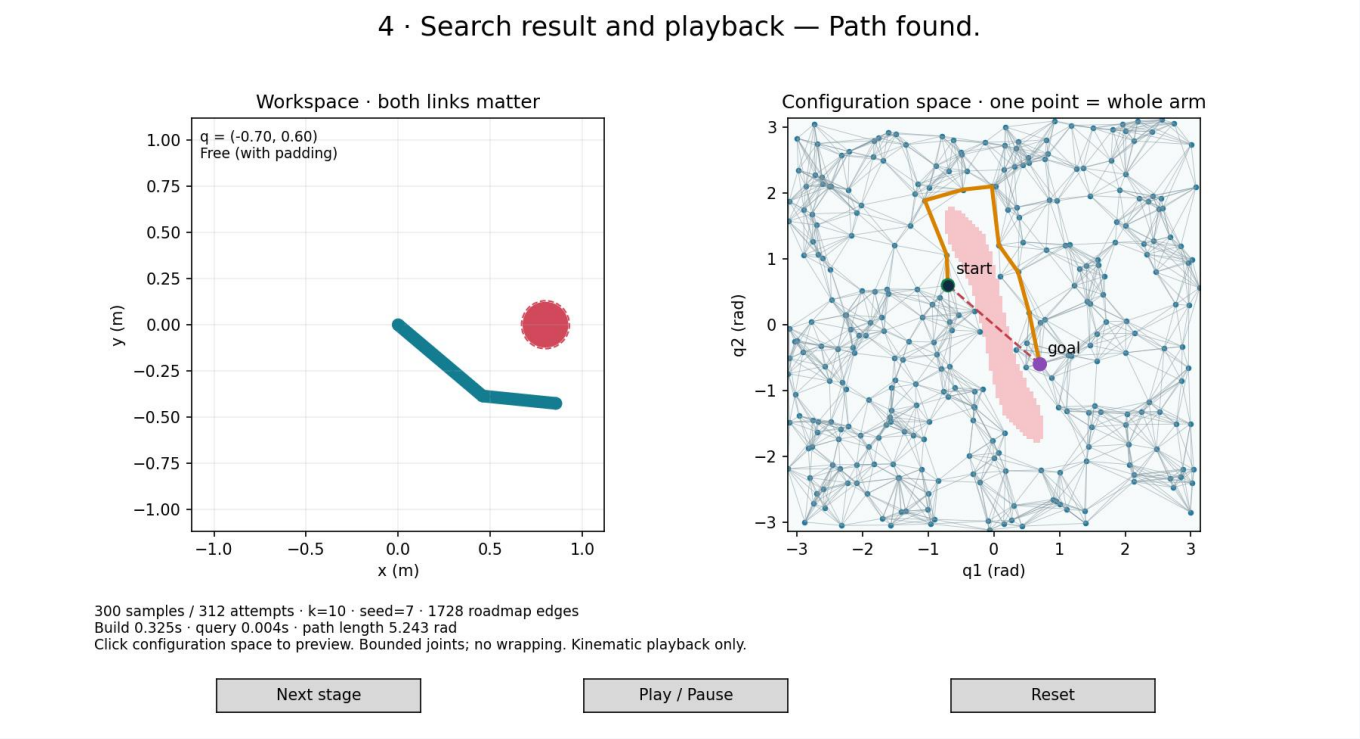
1,728 edges
Path length: 5.243 rad

Both found a route in the documented TA run.

More samples can help; success and better cost are not guaranteed.

Live demo: predict, build, animate

Watch one robot pose and one configuration-space point move together.



5 min · read the code

12 min · reveal + animate

8 min · compare + recap

OPTIONAL / DEEPER DISCUSSION

Collision geometry

Distance to a link segment becomes capsule clearance.

$$u = \mathrm{clamp}(((c-a)\cdot(b-a))/\|b-a\|^2, 0, 1)$$

$$d = \|c - (a + u(b-a))\|$$

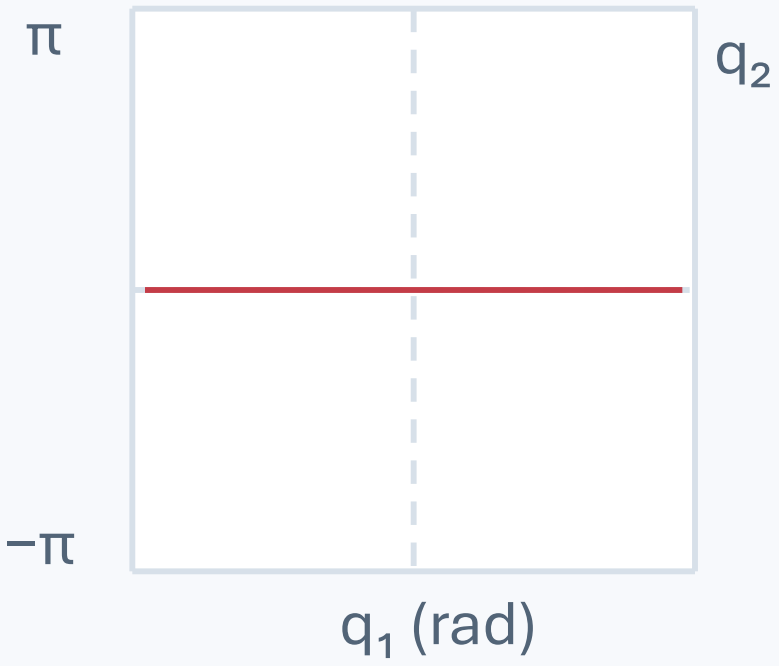
$$\mathrm{clearance} = \min(d_{link1}, d_{link2}) - 0.12 - 0.03$$

Free only when clearance > 0.01 m. Padded contact is blocked.

OPTIONAL / DEEPER DISCUSSION

Topology and interpolation

The demo uses bounded angles; opposite square edges do not wrap.



Boundaries do not wrap.

$$C = [-\pi, \pi] \times [-\pi, \pi]$$

$$q(s) = (1-s)a + sb$$

Unrestricted revolute joints need a periodic model and metric.

OPTIONAL / DEEPER DISCUSSION

PRM construction and queries

Build once; attach actual endpoints to a query copy.

Build roadmap

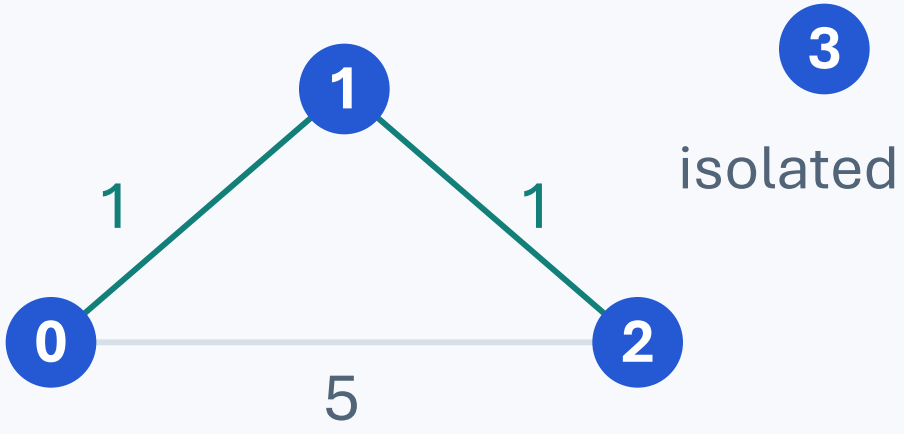
```
sample free q
choose k neighbors
validate local motions
store weighted edges
```

Attach and query

```
copy roadmap
validate + add endpoints
check attachments + direct
Dijkstra; verify route
```

Dijkstra and useful tests

Small known cases expose mistakes before a large demo.



Route [0,1,2], cost 2

Search

Known cost; disconnected; zero length.

Query + geometry

Exact endpoints; copy; blocked midpoint.

OPTIONAL / DEEPER DISCUSSION

Sampling and completeness limits

Finite failure, probabilistic completeness, and optimality are different.

Finite failure

No route in this graph does not prove no physical route.

Completeness

Asymptotic claims require assumptions.

Optimality

Shortest in this graph is not globally shortest.

No completeness or optimality theorem for this fixed-k demo.